

Project Outline

1. Introduction
 - a. Give background on Colorado river and importance to U.S. and Mexico
 - b. Explain how changing weather can cause shifts in water availability
 - c. Research question and hypothesis
 - i. How has decreasing water availability in the Colorado river affected water usage and allocation between the U.S. and Mexico and what role have changing weather conditions played?
2. Data sources and data cleaning
 - a. Introduce USBR Colorado river water usage dataset
 - b. Introduce weather API
 - c. Clean both data sets and merge them into a useful dataset
3. Exploratory data analysis
 - a. Create visualizations and explain why they are relevant
 - i. U.S. vs Mexico water usage
 - ii. Precipitation and water availability
 - iii. Highest uses of water in different levels of water availability
4. Statistical Analysis
 - a. Determine is weather is associated with water availability
 - b. Determine how well weather conditions explain differing water availability
 - c. Compare water usage between U.S. and Mexico using different levels of water availability
5. Discussion and conclusion
 - a. Answer question and conclude whether my hypothesis was correct or not
 - b. Discuss what should happen in order to allow the Colorado to flow normally